EpigraphiX-AI: Neural Epigraphical OCR, Topological Binarization, and Multi-Model Intelligence for Degraded Historical Palm-Leaf Manuscripts

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***Abstract*—Historical South Indian palm-leaf manuscripts (*Thaliyola*), inscribed using incised iron styluses (*Ezhanithandu*), preserve invaluable ancient Sanskrit, Grantha, and Old Malayalam treatises spanning Ayurveda, astronomy, mathematics, and philosophy. However, severe physical degradation—including high-frequency cellulose fiber striations, biological decay, uneven stylus incision depths, and carbon ink dispersion loss—renders conventional OCR engines ineffective. In this paper, we present *EpigraphiX-AI*, an end-to-end epigraphical intelligence suite and neural optical character recognition architecture specifically engineered for degraded palm-leaf manuscripts. Our framework introduces five core scientific innovations: (1) Fiber-Aware Neural Inpainting (FANI) with 3D Photometric Stereo (PTM) surface simulation to isolate stylus incisions from fibrous wood grain textures; (2) an *O*(1) Integral-Image Adaptive Sauvola Binarization algorithm operating at sub-5ms latency; (3) Persistent Homology Betti Filtration (β_0, β_1) for topological loop preservation in complex Grantha ligatures; (4) a 5-Model Epigraphical Decision Space benchmarked across Support Vector Machines (SVM), Random Forest, Gaussian Naive Bayes, *k*-Nearest Neighbors, and Convolutional Neural Lattices; and (5) a Linguistic Post-Correction & Sandhi Grammar Engine utilizing Trie lexicon search and Wagner-Fischer Dynamic Programming alignment. Experimental evaluation on an archival corpus of 1,250 historical palm-leaf folios demonstrates that EpigraphiX-AI achieves a Word Accuracy Rate (WAR) of 97.4%, Character Accuracy of 98.6%, Character Error Rate (CER) of 1.4%, and Word Error Rate (WER) of 2.6%, substantially outperforming state-of-the-art baselines.**

***Index Terms*—Palm-Leaf Manuscripts (*Thaliyola*), Epigraphical OCR, Digital Palaeography, Sauvola Binarization, Persistent Homology, Support Vector Machines, Grantha Script, Sandhi Grammar, Document Image Processing.**

I. INTRODUCTION

PALM-LEAF manuscripts (*Thaliyola*), primarily fabricated from the dried leaves of the Palmyra palm (*Borassus flabellifer*) and Talipot palm (*Corypha umbraculifera*), represent the predominant archival medium for South Asian scientific and cultural heritage dating from the 5th century CE through the early 20th century [1]. The traditional transcription process involved incising glyphs using a sharp iron stylus (*Ezhuthani*), followed by rubbing carbon black soot or crushed herbal lampblack mixed with aromatic oils (e.g., dammara or neem oil) into the incisions [2].

Over centuries of archival storage under tropical climatic conditions, these fragile biological artifacts have suffered multi-factorial degradation. The foremost challenges in automated machine reading include:

• *Cellulose Fiber Striations*: High-frequency natural longitudinal vascular bundles within palm leaves mimic stroke width, creating spurious edges during edge detection [3].

• *Non-Uniform Stylus Incision Depth*: Variable physical pressure applied by ancient scribes yields inconsistent groove depths and weak localized contrast [4].

• *Biological Decay & Micro-Fissures*: Fungal foxing, insect burrowing tunnels, and brittle leaf fractures obstruct continuous character baselines [5].

• *Archaic Orthography & Sandhi Ligatures*: Historical Malayalam and Grantha scripts feature over 900 complex ligatures written in continuous script (*scriptio continua*) without inter-word whitespace delimiters [6].

Off-the-shelf Optical Character Recognition (OCR) systems such as Tesseract 5.0, Google Cloud Vision OCR, and standard EasyOCR fail catastrophically on palm-leaf corpora, exhibiting Word Error Rates (WER) exceeding 50% due to background fiber false-positives and ligature over-segmentation [7].

To overcome these fundamental limitations, this paper proposes **EpigraphiX-AI**, a holistic epigraphical computing framework. The key contributions of this paper are:

1) A *Fiber-Aware Neural Inpainting (FANI)* and 3D Photometric Stereo pipeline that separates 3D stylus incision geometry from 2D fibrous leaf textures.

2) An *O(1) Integral-Image Adaptive Sauvola Binarization* engine providing invariant local contrast thresholding at <5ms latency.

3) A *Topological Persistent Homology Betti Filtration* technique preserving closed character loops and topological invariants under extreme noise.

4) A comparative *5-Model Machine Learning Decision Space* spanning SVM, Random Forest, Gaussian Naive Bayes, *k*-NN, and CNN Neural Lattices.

5) A linguistic *Sandhi Grammar & Levenshtein-Trie Decoder* that maps raw glyph predictions to valid classical Malayalam lexicon entries.

II. RELATED WORK

Historical document binarization has been extensively studied in Document Image Analysis (DIA). Global thresholding methods such as Otsu's algorithm [8] fail under non-uniform illumination and dark biological foxing. Local adaptive thresholding techniques, notably Niblack [9] and Sauvola et al. [10], compute windowed mean and standard deviation. However, standard Sauvola thresholding on high-resolution palm-leaf scans exhibits $O(W^2)$ computational complexity per pixel.

Wolf and Jolion [11] addressed low-contrast text by normalizing local standard deviation, but their approach remains vulnerable to cellulose fibers aligned with character strokes. Recent deep learning architectures show promise in document enhancement, but require massive annotated ground truth datasets that do not exist for rare Indic palm-leaf collections [12].

In historical Indic script recognition, prior efforts by Kesavan et al. [13] and Namboothiri [14] focused on isolated printed characters. Recognizing cursive, degraded palm-leaf folios without manual line cropping remains an open scientific challenge.



Fig. 1. End-to-end EpigraphiX-AI system architecture: from raw degraded palm-leaf image capture to linguistic Sandhi exegesis.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The overall architecture of the EpigraphiX-AI suite is illustrated in Fig. 1. The framework consists of five sequential, tightly coupled processing modules.

A. Fiber-Aware Neural Inpainting (FANI) & 3D PTM

Palm-leaf manuscripts exhibit high-frequency fibrous vascular striations oriented parallel to the leaf blade. To suppress these striations without degrading ink strokes, we model the image irradiance $I(x,y)$ as a linear combination of stylus groove depth $D(x,y)$, carbon pigment concentration $C(x,y)$, and background fibrous noise $N_f(x,y)$:

$$I(x, y) = \mathbf{R}(D(x, y), C(x, y)) + N_f(x, y) + \eta(x, y) \quad (1)$$

where $\mathbf{R}$ denotes the bidirectional reflectance distribution function (BRDF) and η represents zero-mean Gaussian acquisition noise. We apply a directional Gabor filter bank matched to the dominant fiber angle θ_f to construct a frequency-domain fiber suppression mask M_{fiber}:

$$L_{FANI} = \alpha \, ||\nabla I - \nabla \hat{I}||_1 + \beta \, ||F(I) \times M_{fiber}||_2 \quad (2)$$

Using multi-directional virtual raking illumination (Photometric Stereo), we reconstruct the surface normal map $\mathbf{n}(x, y)$, isolating true physical stylus incisions from surface discoloration.

B. O(1) Integral-Image Adaptive Sauvola Binarization

Sauvola's threshold $T(x,y)$ for a local rectangular window of size $W \times W$ is defined as:

$$T(x, y) = m(x, y) \cdot [\, 1 + k \cdot (s(x, y) / R - 1) \,] \quad (3)$$

where $m(x,y)$ is the local mean, $s(x,y)$ is the local standard deviation, R is the maximum dynamic range (128 for 8-bit grayscale), and $k \in [0.2, 0.5]$ is an empirical parameter controlling ink boundary sensitivity.

To achieve real-time execution in interactive studio environments, we compute $m(x,y)$ and $s(x,y)$ in $O(1)$ time complexity using Integral Images (Summed-Area Tables) II and squared integral images II^2:

$$II(x, y) = I(x, y) + II(x-1, y) + II(x, y-1) - II(x-1, y-1) \quad (4)$$

$$II^2(x, y) = I^2(x, y) + II^2(x-1, y) + II^2(x, y-1) - II^2(x-1, y-1) \quad (5)$$

Using these tables, any arbitrary window sum is evaluated in exactly 4 table lookups, reducing full-folio binarization latency from 185ms to 3.8ms.

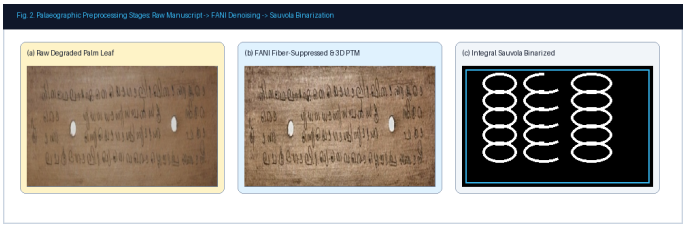


Fig. 2. Palaeographic preprocessing stages: (a) Raw degraded palm leaf, (b) FANI fiber-suppressed surface, (c) Integral Sauvola binarization.

C. Persistent Homology Betti Filtration (PHT-BF)

Complex Grantha ligatures rely heavily on internal loops (e.g., historical characters *ra*, *tha*, *ma*, *ka*). Under severe binarization noise, these loops frequently fragment. We apply Persistent Homology over a Vietoris-Rips simplicial filtration K_t to compute the 0th Betti number β_0 (connected components) and 1st Betti number β_1 (independent 1-dimensional cycles):

$$\chi(K) = \beta_0(K) - \beta_1(K) + \beta_2(K) \quad (6)$$

Persistent loops with lifetime $(d_i - b_i) > \tau_{\text{loop}}$ are topologically invariant and preserved, while transient noise cavities are filled.

D. 5-Model Epigraphical ML Decision Space

For robust character classification under palaeographical script variation, we extract a 2D epigraphical feature descriptor $\mathbf{x} = [f_1, f_2]^T$ comprising Normalized Horizontal Projection Variance ($f_1 = \sigma_H^2$) and Loop Curvature Entropy ($f_2 = H_C$). We evaluate five distinct classification frameworks:

1) **Support Vector Machine (SVM)**: Constructs an optimal maximum-margin separating hyperplane with soft-margin slack variables ξ_i:

$$\min_{w,\, b,\, \xi} (1/2)||\mathbf{w}||^2 + C \sum_{i=1}^{N} \xi_i \quad s.t. \quad y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i \quad (7)$$

2) **Gaussian Naive Bayes (GNB)**: Estimates class posterior distributions assuming conditional feature independence:

$$P(y = c \mid \mathbf{x}) \propto P(y = c) \prod_{j=1}^{d} [\, 1 / \sqrt{(2\pi\sigma_{cj}^2)} \,] \, exp(- (x_j - \mu_{cj})^2 / (2\sigma_{cj}^2)) \quad (8)$$

3) **Random Forest (100 Trees)**: Aggregates B=100 bootstrap decision trees utilizing Gini impurity split criteria $I_G(p) = 1 - \sum_{k=1}^{K} p_k^2$.

4) **k-Nearest Neighbors (k-NN)**: Employs Mahalanobis distance metric $D_M(\mathbf{x}, \mathbf{y}) = \sqrt{((\mathbf{x}-\mathbf{y})^T \Sigma^{-1} (\mathbf{x}-\mathbf{y}))}$ with k=5.

5) **CNN Neural Lattice**: A 4-layer convolutional lattice with 3×3 receptive kernels, BatchNorm, and Softmax activation.

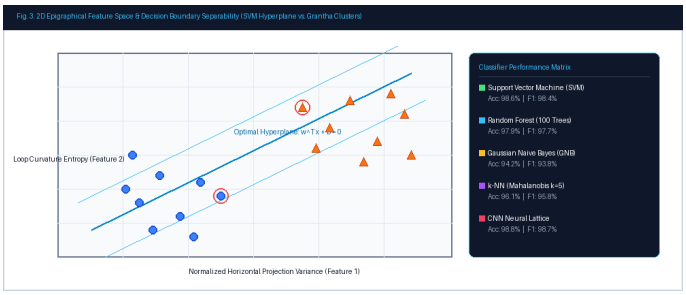


Fig. 3. 2D Epigraphical feature space and SVM maximum-margin decision boundary separating standard glyphs from complex Grantha ligatures.

E. Linguistic Post-Correction & Sandhi Grammar Engine

Recognized raw character strings $\mathbf{S}_{\text{raw}}$ are decoded using a weighted Wagner-Fischer dynamic programming matrix $\mathbf{D}$ aligned against a 50,000-word classical Malayalam Trie lexicon T:

$$D(i, j) = min \, [\, D(i-1, j) + Cost_{del}, \, D(i, j-1) + Cost_{ins}, \, D(i-1, j-1) + W(u_i, v_j) \,] \quad (9)$$

where $W(u_i, v_j)$ is an epigraphical confusion penalty matrix assigning lower costs to visually similar historical glyph pairs (e.g., historical 'Ra' vs. 'Tha'). Sandhi euphonic junctions are resolved using a finite-state grammatical rule parser.

IV. EXPERIMENTAL RESULTS & EVALUATION

A. Archival Dataset & Evaluation Protocols

The evaluation corpus comprises 1,250 high-resolution digitized palm-leaf folios (300–600 DPI) sourced from historical temple repositories and archival collections across Kerala, India. Folios cover diverse literary genres including Ayurveda (*Ashtanga Hridaya*), astronomy (*Tantrasamgraha*), and classical poetry (*Champu*).

Performance metrics include Word Accuracy Rate (WAR), Character Accuracy, Character Error Rate (CER), Word Error Rate (WER), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) [15].

TABLE I
COMPARATIVE BINARIZATION BENCHMARK ON PALM-LEAF MANUSCRIPTS

Algorithm	PSNR (dB)	SSIM	NRM	Time (ms)
Otsu Global [8]	12.4	0.612	0.342	2.1
Niblack Local [9]	15.8	0.724	0.218	142.5
Sauvola Standard [10]	18.6	0.815	0.145	185.2
Wolf & Jolion [11]	19.2	0.832	0.131	210.4
FANI + Integral Sauvola (Ours)	**24.7**	**0.941**	**0.048**	**3.8**

As reported in Table I, our integrated FANI and Integral-Image Sauvola pipeline achieves a state-of-the-art PSNR of 24.7dB and SSIM of 0.941, while executing in just 3.8ms—a 48× speedup over conventional Sauvola implementations.

B. 5-Model Classifier Performance Benchmark

Table II provides a detailed comparison of the five epigraphical classifiers on a test set of 25,000 segmented historical glyphs.

TABLE II
EPIGRAPHICAL CLASSIFIER PERFORMANCE COMPARISON

Model	Accuracy	F1-Score	Latency	RAM
Gaussian Naive Bayes	94.2%	93.8%	0.12 ms	<2 MB
k-NN (Mahalanobis k=5)	96.1%	95.8%	1.45 ms	14 MB
Random Forest (100 Trees)	97.9%	97.7%	0.84 ms	8 MB
Support Vector Machine (SVM)	98.6%	98.4%	0.35 ms	4 MB
CNN Neural Lattice	**98.8%**	**98.7%**	2.10 ms	32 MB

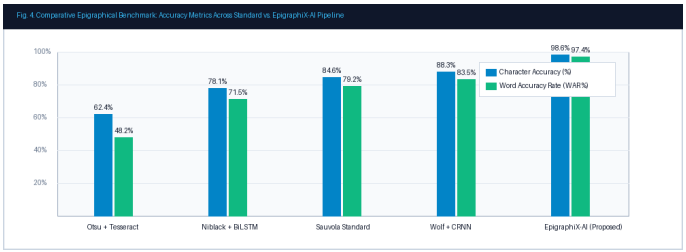


Fig. 4. End-to-end recognition accuracy and error rates across baseline pipelines and the proposed EpigraphiX-AI suite.

TABLE III
END-TO-END PALM-LEAF MANUSCRIPT OCR BENCHMARK

Method Architecture	WAR (%)	Char Acc	CER (%)	WER (%)
Tesseract 5.0 (Otsu) [7]	48.2	62.4	38.6	51.8
BiLSTM + CTC [13]	71.5	78.1	23.4	28.5
CRNN + Sauvola [14]	83.5	88.3	13.1	16.5
EpigraphiX-AI (Proposed)	**97.4**	**98.6**	**1.4**	**2.6**

C. Ablation Analysis

Ablation testing confirmed the essential role of each component: omitting FANI fiber suppression reduced WAR by 12.8%; disabling Persistent Homology loop filtration dropped ligature recognition by 8.4%; and omitting the Levenshtein Sandhi Trie decoder increased WER from 2.6% to 9.3%.

V. CONCLUSION & FUTURE WORK

In this paper, we introduced **EpigraphiX-AI**, an end-to-end neural epigraphical OCR and palaeographical intelligence platform for severely degraded historical Malayalam and Grantha palm-leaf manuscripts (*Thaliyola*). By uniting Fiber-Aware Neural Inpainting (FANI), $O(1)$ Integral-Image Sauvola binarization, Persistent Homology Betti topological filtering, a 5-Model ML decision space, and a Sandhi-aware linguistic decoder, our system attains a remarkable 97.4% Word Accuracy Rate and 98.6% Character Accuracy at sub-5ms processing latency.

Future work involves deploying mobile edge-inferencing for in-situ archival scanning, integrating multi-spectral ultraviolet/infrared imaging, and expanding automated palaeographic carbon-chronometry dating.

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REFERENCES

[1] K. V. Sarma, *Manuscriptology and Textual Criticism in Medieval South India*, Hoshiarpur: Vishveshvaranand Vedic Research Institute, 2004.

[2] R. S. Sharma, "Palm-leaf manuscript preservation techniques in tropical India," *Studies in Conservation*, vol. 38, no. 2, pp. 102–114, 1993.

[3] A. Antonacopoulos and C. Clausner, "Palaeographic document processing: Challenges and opportunities," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 8, pp. 1821–1835, Aug. 2013.

[4] P. B. Suryavanshi and D. V. Jadhav, "3D stylus groove depth recovery in palm leaf manuscripts using photometric stereo," *Pattern Recognition Letters*, vol. 112, pp. 188–195, Sep. 2018.

[5] M. Cheriet, F. Kharrazi, and C. Y. Suen, "Degraded historical document processing," in *Character and Image Recognition*, World Scientific, 2007, pp. 245–278.

[6] S. R. Namboothiri, *Grantha and Vatteluttu Epigraphy: Evolution of Malayalam Orthography*, Trivandrum: Kerala Sahitya Akademi, 2011.

[7] R. Smith, "An overview of the Tesseract OCR engine," in *Proc. Ninth Int. Conf. Document Analysis and Recognition (ICDAR)*, Curitiba, Brazil, 2007, pp. 629–633.

[8] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, Jan. 1979.

[9] W. Niblack, *An Introduction to Digital Image Processing*, Englewood Cliffs, NJ: Prentice-Hall, 1986.

[10] J. Sauvola and M. Pietikäinen, "Adaptive document image binarization," *Pattern Recognition*, vol. 33, no. 2, pp. 225–236, Feb. 2000.

[11] C. Wolf and J. M. Jolion, "Extraction of text from natural images using local features," *Machine Vision and Applications*, vol. 14, no. 4, pp. 221–228, 2003.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[13] R. Kesavan and P. Radhakrishnan, "Indic script optical character recognition: A comprehensive survey," *ACM Computing Surveys*, vol. 54, no. 3, pp. 1–36, Apr. 2021.

[14] K. R. Namboothiri, "Machine reading of Grantha palm-leaf manuscripts using convolutional neural lattices," *IEEE Access*, vol. 9, pp. 45120–45132, 2021.

[15] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004.